

Linear Algebra (LAG1) — Complete Test

Practice Questions with Answers and Explanations

Question: 77

Language: English

How to use this book

This collection contains all 77 questions of the test “Linear Algebra (LAG1) — Complete Test”. Every question shows all answer choices, the correct answer highlighted in green, and a detailed explanation below it. Try to answer on your own first, then check yourself. The answer key on the last page allows quick self-testing.

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1. In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the sphere S defined by the equation $x^2+y^2+z^2-2x+2z=0$. S intersects the line given by $(x,y,z)=(0,t,0)$ in two distinct points S passes through the point $(1,-1,1)$ the center of S has coordinates $(2,1,-1)$ S has radius $\sqrt{2}$

- A. S intersects the line given by $(x,y,z)=(0,t,0)$ in two distinct points
- B. S passes through the point $(1,-1,1)$
- C. the center of S has coordinates $(2,1,-1)$
- D. S has radius $\sqrt{2}$ ✓

EXPLANATION

D, radius = $\sqrt{(1)^2 + (-1)^2 + (1)^2} = \sqrt{3} \rightarrow (x-1)^2+(y+1)^2+(z-1)^2=3$, so radius $\sqrt{2}$ is wrong.

2. Let S be a linear system of 4 equations and 3 variables, and let $M=(A|B)$ be the complete matrix associated to S if $\det(M) \neq 0$, which one of the following statements is certainly true

- A. S admits infinite solutions
- B. S does not have any solutions ✓
- C. S admits only one solution
- D. the zero vector is a solution of S

EXPLANATION

C) S admits only one solution, because if $\det(M) \neq 0$, the system has a unique solution (invertible matrix).

3. Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant. Find the right statement.

- A. A is not positive or negative definite. ✓
- B. A is positive or negative semi-definite.
- C. 0 is not an eigenvalue of A .
- D. None of the other statements is correct.

EXPLANATION

A symmetric matrix with zero trace and determinant cannot be definite (eigenvalues sum to 0, product is 0).

4. For all $t \in \mathbb{R}$, consider the points $P = (1,2,3)$, $Q = (0,0,t)$, $R = (1,-1,t)$, $S = (1,1,t)$. Which of the following statements is true?

- A. Points P, Q, R, S are vertices of a parallelogram for all values of t .
- B. Points P, Q, R, S are coplanar for a unique value of t . ✓
- C. Points P, Q, R, S are vertices of a rectangle for all values of t .
- D. Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t .

EXPLANATION

Points are coplanar when the scalar triple product of vectors PQ, PR, PS is zero, which happens only for a unique t .

5. Let us consider the quadratic form defined by $f(x,y) = (x,y)A(x,y) = x^2 - 3xy + 8y^2$, where A is a 2×2 real symmetric matrix. Which of the following statements is true? (a) If $xy \neq 0$, then $f(x,y) > 0$ (b) The curve defined by $f(x,y) = -1$ has at least a point w (c) The determinant of A is negative. (d) There exists $(a,b) \in \mathbb{R}^2$ such that $f(a,b) < 0$.

- A. If $xy \neq 0$, then $f(x,y) > 0$ ✓
- B. The curve defined by $f(x,y) = -1$ has at least a point w
- C. The determinant of A is negative.
- D. There exists $(a,b) \in \mathbb{R}^2$ such that $f(a,b) < 0$.

EXPLANATION

$f(x,y) = x^2 - 3xy + 8y^2 = (x - (3/2)y)^2 + (23/4)y^2 > 0$ for all $(x,y) \neq (0,0)$.

6. Let us consider the linear system with 4 variables depending on the real parameter k : $\{ x - t = 0, -x + 2y + 2z + t = 4, y + z = k \}$. Find the right statement.

- A. If $k = 2$ the system admits ∞^2 solutions. ✓
- B. If $k = 2$ the system admits ∞^1 solutions.
- C. If $k \neq 2$ the system admits only one solution.
- D. For any $k \in \mathbb{R}$ the system does not admit any solution.

EXPLANATION

For $k=2$, the system has infinitely many solutions (∞^2) as the rank of the coefficient matrix is 2 and the augmented matrix is 3.

7. Let us consider the linear system with 4 variables depending on the real parameter k : $\{ x - t = 0, -x + 2y + 2z + t = 4, y + z = k \}$. Find the right statement.

- A. (a) If $k = 2$ the system admits ∞^2 solutions. ✓
- B. (b) If $k = 2$ the system admits ∞^1 solutions.
- C. (c) If $k \neq 2$ the system admits only one solution.
- D. (d) For any $k \in \mathbb{R}$ the system does not admit any solution.

EXPLANATION

For $k=2$, the system has infinitely many solutions (∞^2) as the rank of the coefficient matrix is 2 and the augmented matrix is 3.

8. Let us consider the linear system with 4 variables depending on the real parameter $k \in \mathbb{R}$

$$\begin{aligned}x - t &= 0 \\ -x + 2y + 2z + t &= 4 \\ y + z &= k\end{aligned}$$

Find the right statement.

- A. (a) If $k = 2$ the system admits ∞^2 solutions. ✓
- B. (b) If $k = 2$ the system admits ∞^1 solutions.
- C. (c) If $k \neq 2$ the system admits only one solution.
- D. (d) For any $k \in \mathbb{R}$ the system does not admit any solution.

EXPLANATION

For $k=2$, the system has infinitely many solutions (∞^2) as the rank of the coefficient matrix is 2 and the augmented matrix is 3.

9. Let r be the line $\{x=2+t, y=2-t, z=t\}$. Which of the following statements is true? (a) There exists a unique plane containing r and parallel to vector, (b) There exists a unique plane containing r and the point $(2,2,0)$, (c) There exists a unique plane containing r and the points $(2,2,0)$ and $(0,4,0)$, (d) There exists a unique plane containing r and orthogonal to vector

- A. (a) There exists a unique plane containing r and parallel to vector
- B. (b) There exists a unique plane containing r and the point $(2,2,0)$
- C. (c) There exists a unique plane containing r and the points $(2,2,0)$ and $(0,4,0)$ ✓
- D. (d) There exists a unique plane containing r and orthogonal to vector

EXPLANATION

C, two points on r are $(2,2,0)$ and $(3,1,1)$, defining a unique plane.

10. Let r be the line $\{x=2+t, y=2-t, z=t\}$. Which of the following statements is true? (a) There exists a unique plane containing r and parallel to vector $v=(1,-1,1)$ (b) There exists a unique plane containing r and the point $(2,2,0)$ (c) There exists a unique plane containing r and the points $(2,2,0)$ and $(0,4,0)$ (d) There exists a unique plane containing r and orthogonal to vector $v=(1,-1,1)$

- A. There exists a unique plane containing r and parallel to vector $v=(1,-1,1)$
- B. There exists a unique plane containing r and the point $(2,2,0)$
- C. There exists a unique plane containing r and the points $(2,2,0)$ and $(0,4,0)$ ✓
- D. There exists a unique plane containing r and orthogonal to vector $v=(1,-1,1)$

EXPLANATION

The points (2,2,0) and (0,4,0) are not on the line r , so they define a unique plane with r .

11. In the Euclidean plane with fixed Cartesian system Oxy , let C be the conic defined by the equation $7x^2 + y^2 + 2x + y = 0$. Which of the following statements is correct?

- A. C is a hyperbola
- B. C is degenerate
- C. C is an ellipse ✓
- D. C is a parabola

EXPLANATION

C , $7x^2 + y^2 + 2x + y = 0 \rightarrow (x+1/7)^2 + (y+1/2)^2 = \text{constant}$, so it's an ellipse.

12. 1. If the linear system $Ax=b$ with 5 equations and 6 variables has ∞^m solutions with $m \leq 3$ then necessarily

- (a) $\text{rank}(A/B) \geq 2$
- (b) $\text{rank}(A) \geq 2$
- (c) $\text{rank}(A) = 2$
- (d) $\text{rank}(A) \leq 2$

2. Let i, j, k be the versors of the coordinate axes of $\mathbb{R}^3$. For every $a \in \mathbb{R}$ let us consider the vectors $v = aj + ak$, $v = 2i + j$, $w = E$

- (a) There exists a unique $a \in \mathbb{R}$ that vectors are coplanar
- (b) There exists a unique $a \in \mathbb{R}$ that vectors are linearly independent
- (c) For every $a \in \mathbb{R}$ vectors are coplanar
- (d) For every $a \in \mathbb{R}$ vectors are linearly independent

3. In the Euclidean 3-dimensional space with a fixed cartesian system, let us consider $C_1: x^2 + y^2 + z^2 - 1 = 0$, $C_2: x^2 + y^2 + z^2 - 7z - x - 2 = 0$ having radii respectively

- (a) $R_1 = R_2$
- (b) $R_1 < R_2$
- (c) C_1 consists of only one point
- (d) $R_1 > R_2$

4. In $\mathbb{R}^4$ let us consider the vector subspace $V = (x, y, z, w) \mid x + y + z + w = 0, x - 3y = 0$ True?

- (a) V is the vector subspace generated by the vectors (3,1,0,-4) and (0,0,1,-1)
- (b) A basis of V is formed by 3 vectors
- (c) V is the vector subspace generated by the vectors (1,1,1,1) and (0,0,1,-1)
- (d) V is subspace of dimension 1

- A. (a) $\text{rank}(A/B) \geq 2$
- B. (b) $\text{rank}(A) \geq 2$ ✓
- C. (c) $\text{rank}(A) = 2$
- D. (d) $\text{rank}(A) \leq 2$

EXPLANATION

1. $\text{rank}(A) \geq 2$ since ∞^m solutions ($m \leq 3$) implies $\text{rank}(A) < 6$ and $\text{trace}(A)=0$ implies $\text{rank}(A) \geq 2$, Q2: 2. A is symmetric with zero trace and $\det(A)=0$, so it cannot be definite, Q3: 3. The quadratic form $x^2 - 3xy + 8y^2$ is positive definite (discriminant $9-32 < 0$, leading coefficient > 0), Q4: 4. The plane is uniquely determined by the line and two non-collinear points, Q5: 5. A homogeneous system with non-zero solutions implies $\det(A)=0$, Q6: 6. The plane orthogonal to r at $(2,2,0)$ is unique, Q7: 7. Same as Q2, Q8: 8. Same as Q2, Q9: 9. The set is closed under addition and scalar multiplication ($a+b=0 \Rightarrow (a'+b')+(a''+b'')=0$), Q10: 10. $\det(A)=1$, and $A \cdot A^{-1}=I$ verifies the inverse, Q11: 11. The matrix of f has eigenvalues 0, -1, -2 with eigenvectors $(1,0,1)$, $(0,1,0)$, $(1,0,-1)$, Q12: 12. Same as Q5, Q13: 13. Same as Q2, Q14: 14. The volume of the tetrahedron is $|\det([Q-P, R-P, S-P])|/6 = |t-3|/6$, Q15: 15. The kernel has dimension 1 since the image has dimension 2 (rank of the matrix is 2), Q16: 16. $\dim(U \cap V) = \dim(U) + \dim(V) - \dim(U+V) \geq 3+4-5 = 2$, Q17: 17. $(2,0)$ is an eigenvector with eigenvalue 3, Q18: 18. $\text{rank}(A) \leq 4$ since there are ∞^m solutions ($m \geq 1$), Q19: 19. The characteristic polynomial has degree 4, so $\dim(\text{Im}(f))=4$, Q20: 20. The quadratic form has eigenvalues $2.5 \pm \sqrt{0.25} > 0$, so it's positive definite, Q21: 21. Both spheres have radius $\sqrt{(1+1/4)} = \sqrt{5/2}$, Q22: 22. $\det(A) = (-1)^5 \det(U) = (-1)^2 \cdot 6 = 6$, Q23: 23. Rounding 0.13819 to 3 digits gives 0.138, Q24: 24. $A \cdot w^m \approx \lambda_1 \cdot w^m \Rightarrow \lambda_1^m \approx 3$, Q25: 25. spline computes and evaluates the not-a-knot spline, Q26: 26. The points are never coplanar (volume of tetrahedron is always 2), Q27: 27. Same as Q16, Q28: 28. Same as Q17, Q29: 29. Same as Q18, Q30: 30. Cholesky factorization of a symmetric positive definite matrix is unique, Q31: 31. Same as Q25, Q32: 32. $513.21 \cdot 20.135 = 10333.48335 = 0.1033348335 \cdot 10^5$, Q33: 33. Completing the square gives $(x^2) + (y-5)^2 = 0$, so C is a single point, Q34: 34. The distance from the sphere's center to π_2 is greater than to π_1 , so $R_2 < R_1$, Q35: 35. The kernel has dimension 1 since the image has dimension 2 (rank of the matrix is 2), Q36: 36. The dot product $(P_1 P_2) \cdot (P_1 P_3) = 0$ has a real solution, Q37: 37. A has eigenvalues 0, 0, 10 (not positive definite),

13. In the euclidean 3-dimensional space with fixed cartesian system: $x^2 + y^2 + z^2 - 2y + 2z = 0$

- A. $x^2 + (y-1)^2 + (z+1)^2 = 2$ ✓
- B. $x^2 + (y-1)^2 + (z+1)^2 = -2$
- C. $x^2 + (y-1)^2 + (z+1)^2 = 1$
- D. $x^2 + (y-1)^2 + (z+1)^2 = 0$

EXPLANATION

Completing the square gives $x^2 + (y-1)^2 + (z+1)^2 = 2$.

14. We recall that an endomorphism $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is said to be simple if there exists a basis of V of eigenvectors of f . Let us consider the endomorphism $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $f(x,y,z) = (-x + z, -2y, x - z)$. Which of the following statements is correct?

- A. All the eigenspaces of f have dimension 1.
- B. The image of f has dimension 1.
- C. f is simple. ✓
- D. f is an isomorphism.

EXPLANATION

C , f has eigenvalues 1, -2, -1 with eigenvectors $(1,0,1)$, $(0,1,0)$, $(1,0,-1)$, forming a basis.

15. Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zero solution and find the correct statement.

- A.** $\det(A) = 0$. ✓
- B.** $\text{rank}(A) = n$.
- C.** A is an invertible matrix.
- D.** None of the previous statements is correct.

EXPLANATION

A homogeneous system with a non-zero solution implies $\det(A) = 0$.

16. Let us consider the line $r: \{x = 2 + t, y = 2 - t, z = t\}$. Which of the following statements is true?

- A.** There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r . ✓
- B.** There exists a unique plane containing the point $(2, 2, 0)$ and parallel to r .
- C.** There exists a unique plane containing r and the point $(2, 2, 0)$.
- D.** There exists a unique plane containing r and orthogonal to the vector $v = (1, -1, 1)$.

EXPLANATION

A plane orthogonal to line r at $(2, 2, 0)$ is unique.

17. Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant. Find the right statement.

- A.** A is not positive or negative definite. ✓
- B.** A is positive or negative semi-definite.
- C.** 0 is not an eigenvalue of A .
- D.** None of the other statements is correct.

EXPLANATION

A symmetric matrix with zero trace and determinant cannot be definite (eigenvalues sum to 0 , product is 0).

18. Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant. Find the right statement.

- A.** A is not positive or negative definite. ✓
- B.** A is positive or negative semi-definite.
- C.** 0 is not an eigenvalue of A .
- D.** None of the other statements is correct.

EXPLANATION

A symmetric matrix with zero trace and determinant cannot be definite (eigenvalues sum to 0 , product is 0).

19. For all $t \in \mathbb{R}$, consider the points $P = (1, 2, 3)$, $Q = (0, 0, t)$, $R = (1, -1, t)$, $S = (1, 1, t)$. Which of the following statements is true?

- A.** Points P, Q, R, S are vertices of a parallelogram for all values of t .
- B.** Points P, Q, R, S are coplanar for a unique value of t . ✓
- C.** Points P, Q, R, S are vertices of a rectangle for all values of t .
- D.** Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t .

EXPLANATION

Points are coplanar when the scalar triple product of vectors PQ, PR, PS is zero, which happens only for a unique t .

20. Let $R_2[x]$ be the vector space of real polynomials of degree ≤ 2 . Let $f: R_2[x] \rightarrow \mathbb{R}^3$ be a linear map such that $f(x^2) = (0,0,1)$, $f(x) = (0,1,0)$, $f(1) = (0,3,3)$. Which one of the following statements is true?

- A. $f(3x - 1) = (0,0,3)$
- B. $\dim(\text{Ker } f) = 1$ ✓
- C. $\dim(\text{Ker } f) = 2$
- D. f is surjective

EXPLANATION

$\dim(\text{Ker } f) = 1$ because the image of f has dimension 2 (spanned by $(0,1,0)$ and $(0,0,1)$), so by rank-nullity theorem, $\dim(\text{Ker } f) = 3 - 2 = 1$.

21. Let $R_3[x]$ be the vector space of polynomials in the variable x , of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $R_3[x]$ is a vector subspace?

- A. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b = 0\}$. ✓
- B. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ad - bc = 0\}$.
- C. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b + c + d = 1\}$.
- D. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ac = 0\}$.

EXPLANATION

The set is closed under addition and scalar multiplication, satisfying subspace criteria.

22. Let $R_3[x]$ be the vector space of polynomials in the variable x , of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $R_3[x]$ is a vector subspace?

- A. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b = 0\}$. ✓
- B. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ad - bc = 0\}$.
- C. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b + c + d = 1\}$.
- D. The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ac = 0\}$.

EXPLANATION

A, $\{p \mid a+b=0\}$ is closed under addition and scalar multiplication.

23. Consider the matrix $A = \begin{pmatrix} -1 & -1 & 2 & 1 & 0 & 1 & 0 & -1 & 2 \end{pmatrix}$ Find the correct statement.

- A. $A^{-1} = \begin{pmatrix} 1 & 0 & 1 & 2 & -1 & -3 & 1 & -1 & -1 \end{pmatrix}$
- B. $A^{-1} = \begin{pmatrix} 1 & 0 & -1 & -1 & 1 & 2 & -1 & 1 & 1 \end{pmatrix}$
- C. $A^{-1} = \begin{pmatrix} 1 & 0 & -1 & 2 & 2 & 2 & -1 & 1 & 1 \end{pmatrix}$
- D. $A^{-1} = \begin{pmatrix} -1 & 0 & 1 & 2 & 2 & -3 & 1 & 1 & -1 \end{pmatrix}$ ✓

EXPLANATION

$\det(A) = -1$, and $A * A^T$ confirms the correct inverse.

24. We recall that an endomorphism $f : V \rightarrow V$ is said to be simple if there exists a basis of V of eigenvectors of f . Let us consider the endomorphism $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $f(x,y,z) = (-x + z, -2y, x - z)$. Which of the following statements is correct?

- A. All the eigenspaces of f have dimension 1.
- B. The image of f has dimension 1.
- C. f is simple. ✓
- D. f is an isomorphism.

EXPLANATION

The matrix has a basis of eigenvectors, making f simple.

25. Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zero solution and find the correct statement.

- A. $\det(A) = 0$. ✓
- B. $\text{rank}(A) = n$.
- C. A is an invertible matrix.
- D. None of the previous statements is correct.

EXPLANATION

A homogeneous system with a non-zero solution implies $\det(A) = 0$.

26. Let us consider the line $r: \{ x = 2 + t, y = 2 - t, z = t \}$. Which of the following statements is true?

- A. There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r . ✓
- B. There exists a unique plane containing the point $(2, 2, 0)$ and parallel to r .
- C. There exists a unique plane containing r and the point $(2, 2, 0)$.
- D. There exists a unique plane containing r and orthogonal to the vector $v = (1, -1, 1)$.

EXPLANATION

The plane orthogonal to r (direction vector $(1,-1,1)$) through $(2,2,0)$ is unique: $(x-2) - (y-2) + z = 0$.

27. Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant. Find the right statement.

- A. A is not positive or negative definite. ✓
- B. A is positive or negative semi-definite.
- C. 0 is not an eigenvalue of A .
- D. None of the other statements is correct.

EXPLANATION

A symmetric matrix with zero trace and determinant must have both positive and negative eigenvalues, so it's indefinite.

28. Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant. Find the right statement.

- A. A is not positive or negative definite. ✓
- B. A is positive or negative semi-definite.
- C. 0 is not an eigenvalue of A .
- D. None of the other statements is correct.

EXPLANATION

A symmetric matrix with zero trace and determinant cannot be definite (eigenvalues sum to 0, product is 0).

29. For all $t \in \mathbb{R}$, consider the points $P = (1,2,3)$, $Q = (0,0,t)$, $R = (1,-1,t)$, $S = (1,1,t)$. Which of the following statements is true?

- A. Points P, Q, R, S are vertices of a parallelogram for all values of t.
- B. Points P, Q, R, S are coplanar for a unique value of t. ✓
- C. Points P, Q, R, S are vertices of a rectangle for all values of t.
- D. Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t.

EXPLANATION

The points are coplanar only when $t = 3$ (volume of tetrahedron is zero).

30. Let $R_2[x]$ be the vector space of real polynomials of degree ≤ 2 . Let $f: R_2[x] \rightarrow \mathbb{R}^3$ be a linear map such that $f(x^2) = (0,0,1)$, $f(x) = (0,1,0)$, $f(1) = (0,3,3)$. Which one of the following statements is true?

- A. $f(3x-1) = (0,0,3)$
- B. $\dim(\text{Ker } f) = 1$ ✓
- C. $\dim(\text{Ker } f) = 2$
- D. f is surjective

EXPLANATION

$\dim(\text{Ker } f) = 1$ because $\dim(\text{Im } f) = 2$ and $\dim(R_2[x]) = 3$.

31. In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations $C_1: x^2 + y^2 + z^2 - 1 = 0$ and $x + y = 0$, $C_2: x^2 + y^2 + z^2 - 1 = 0$ and $x = 0$, having radius respectively R_1 and R_2 . Which of the following statements is true?

- A. $R_1 > R_2$
- B. C_1 consists of only one point
- C. $R_1 < R_2$
- D. $R_1 = R_2$ ✓

EXPLANATION

D, both C_1 and C_2 are circles of radius 1 (intersection of sphere $x^2+y^2+z^2=1$ with planes $x+y=0$ and $x=0$).

32. In the euclidean space let us consider the points $A = (0,1 + a,1)$, $B = (0,2 + a,1)$, $C = (2,1,0)$, $D = (0,1,0)$. Which of the following statements is true?

- A. There exists only one $a \in \mathbb{R}$ such that the points are collinear.
- B. For every $a \in \mathbb{R}$ the points are not coplanar. ✓
- C. For every $a \in \mathbb{R}$ the points are collinear.
- D. There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

EXPLANATION

The points are never coplanar for any a , as the volume of the tetrahedron they form is always non-zero (2).

33. Let U and V be vector subspaces of $\mathbb{R}^5$ such that $\dim(U) = 3$ and $\dim(V) = 4$. Which one of the following statements is always true?

- A. $U + V = \mathbb{R}^5$
- B. $\dim(U \cap V) \geq 2$ ✓
- C. $\dim(U \cap V) = 1$
- D. $U \subseteq V$

EXPLANATION

$\dim(U \cap V) \geq \dim(U) + \dim(V) - \dim(\mathbb{R}^5) = 3 + 4 - 5 = 2$.

34. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the endomorphism $f(x,y) = (3x - y, 3y)$. Which one of the following statements is true?

- A. f has two distinct eigenvalues.
- B. $(2,0)$ is an eigenvector. ✓
- C. $(0,1)$ is an eigenvector.
- D. f is simple.

EXPLANATION

$(2,0)$ is an eigenvector with eigenvalue 3.

35. If the linear system $AX = B$ with 4 equations and 5 variables has ∞^m solutions with $m \geq 1$, then necessarily

- A. $\text{rank}(A|B) = 4$
- B. $\text{rank}(A) = 4$
- C. $\text{rank}(A) \geq 4$
- D. $\text{rank}(A) \leq 4$ ✓

EXPLANATION

$\text{rank}(A) \leq 4$ because the system has infinitely many solutions ($m \geq 1$).

36. Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - 3)$. Which one of the following statements is true? (a) $\dim(\text{Im}(f)) = 4$. (b) $\dim(\text{Im}(f)) = 3$. (c) $\dim(\text{Ker}(f)) = 2$. (d) $\dim(\text{Im}(f)) = 2$.

- A. $\dim(\text{Im}(f)) = 4$ ✓
- B. $\dim(\text{Im}(f)) = 3$
- C. $\dim(\text{Ker}(f)) = 2$
- D. $\dim(\text{Im}(f)) = 2$

EXPLANATION

$\dim(\text{Im}(f)) = 4$ because the characteristic polynomial has no zero roots.

37. Let us consider the quadratic form defined by $q(x,y) = (x,y)A \begin{pmatrix} x \\ y \end{pmatrix}$ with $A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. Which of the following statements is true? (a) The quadratic form $q(x,y)$ is negative definite. (b) $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus (0,0)$. (c) The matrix A has two eigenvalues of different sign. (d) $q(x,y) = 0$ is the equation of two lines passing through the origin.

- A. The quadratic form $q(x,y)$ is negative definite.
- B. $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus (0,0)$ ✓
- C. The matrix A has two eigenvalues of different sign.
- D. $q(x,y) = 0$ is the equation of two lines passing through the origin.

EXPLANATION

Eigenvalues of A are 1 and 4, so $q(x,y) > 0$ for all $(x,y) \neq (0,0)$.

38. In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the spheres $C_1: x^2 + y^2 + z^2 - 1 = x + y = 0$, $C_2: x^2 + y^2 + z^2 - 2x - 2y + 1 = 0$, having radius respectively R_1 and R_2 . Which of the following statements is true?

- A. $R_1 > R_2$
- B. C_1 consists of only one point
- C. $R_1 < R_2$
- D. $R_1 = R_2$ ✓

EXPLANATION

Both spheres have radius $\sqrt{(1^2 + 1^2 + 0^2)} = \sqrt{2}$.

39. Let P , L and U be the factors of the $PA = LU$ factorization of a given matrix A , and let $s = 2$ be the total number of performed row swaps. If $U = \begin{pmatrix} 2 & 5 & 7 \\ 0 & 1 & 9 \\ 0 & 0 & 3 \end{pmatrix}$ then the determinant of A is equal to:

- A. 2
- B. 6 ✓
- C. -6
- D. -2

EXPLANATION

$\det(A) = (-1)^2 \cdot \det(U) = 1 \cdot 6 = 6$.

40. Consider a floating point arithmetic with $N = 10$, $t = 3$ digit reserved for the mantissa and rounding technique "rounding to even". Let p be the mantissa and $\bar{p}$ the corresponding machine mantissa. If $p = 0.13819$, then:

- A. $\bar{p} = 0.1382$
- B. $\bar{p} = 0.138$ ✓
- C. $\bar{p} = 0.139$
- D. $\bar{p} = 0.1381$

EXPLANATION

Rounding 0.13819 to 3 digits gives 0.138.

41. Let the matrix $A = \begin{pmatrix} 3 & 1 & 4 \\ 5 \end{pmatrix}$ be given, and assume to apply the power method to A . Let $w^{(m)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ be the approximation of the eigenvector corresponding to the eigenvalue of maximum modulus λ_1 at step m . What is the value of $\lambda_1^{(m)}$, that is the approximation of λ_1 at step m ?

- A. 15
- B. 9
- C. 3 ✓
- D. 8

EXPLANATION

$\lambda_1^{(m)} \approx (A w^{(m)})_1 / w_1^{(m)} = 3$.

42. We are given a matrix $A \in \mathbb{R}^n \times n$, symmetric and positive definite, and its Choleski factorization $A = R^T(T)R$. Which one of the following statements is surely correct?

- A. the factorization always exists but it is not unique
- B. the uniqueness of the factorization depends on the rank of matrix A
- C. the factorization does not always exist, but when it exists it is unique
- D. the factorization always exists and it is unique ✓

EXPLANATION

Cholesky factorization of a symmetric positive definite matrix always exists and is unique.

43. Consider the following set of Matlab commands: `x=linspace(-1,1,10); xx=linspace(-1,1); y=sin(πx); z=spline(x,y,xx);` What is their result?

- A. They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- B. They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- C. They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- D. They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$ ✓

EXPLANATION

spline computes and evaluates the not-a-knot cubic spline interpolant.

44. In exact arithmetic (base 10), the result of the multiplication $513.21 * 20.135$ represented in the normalized floating point notation is:

- A. $10.33348337 * 10^3$
- B. $0.10333 * 10^6$
- C. $0.1033348335 * 10^5$ ✓
- D. $1.0333 * 10^4$

EXPLANATION

$513.21 * 20.135 = 10333.48335 = 0.1033348335 * 10^5$.

45. In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations $C_1: x^2 + y^2 + z^2 - 1 = 0$ and $x + y = 0$, $C_2: x^2 + y^2 + z^2 - 1 = 0$ and $x = 0$, having radius respectively R_1 and R_2 . Which of the following statements is true?

- A. $R_1 > R_2$
- B. C_1 consists of only one point
- C. $R_1 < R_2$
- D. $R_1 = R_2$ ✓

EXPLANATION

D, both C_1 and C_2 are circles of radius 1 (intersection of sphere $x^2+y^2+z^2=1$ with planes $x+y=0$ and $x=0$).

46. In the euclidean space let us consider the points $A = (0, 1 + a, 1)$, $B = (0, 2 + a, 1)$, $C = (2, 1, 0)$, $D = (0, 1, 0)$. Which of the following statements is true?

- A. There exists only one $a \in \mathbb{R}$ such that the points are collinear.
- B. For every $a \in \mathbb{R}$ the points are not coplanar. ✓
- C. For every $a \in \mathbb{R}$ the points are collinear.
- D. There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

EXPLANATION

For any a , the points are not coplanar (volume of tetrahedron $\neq 0$).

47. Let U and V be vector subspaces of $\mathbb{R}^5$ such that $\dim(U) = 3$ and $\dim(V) = 4$. Which one of the following statements is always true?

- A. $U + V = \mathbb{R}^5$
- B. $\dim(U \cap V) \geq 2$ ✓
- C. $\dim(U \cap V) = 1$
- D. $U \subseteq V$

EXPLANATION

$\dim(U \cap V) \geq \dim(U) + \dim(V) - \dim(\mathbb{R}^5) = 3 + 4 - 5 = 2$.

48. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$. Which one of the following statements is true?

- A. f has two distinct eigenvalues.
- B. $(2, 0)$ is an eigenvector. ✓
- C. $(0, 1)$ is an eigenvector.
- D. f is simple.

EXPLANATION

$(2, 0)$ is an eigenvector with eigenvalue 3.

49. If the linear system $AX = B$ with 4 equations and 5 variables has ∞^m solutions with $m \geq 1$, then necessarily

- A. $\text{rank}(A|B) = 4$
- B. $\text{rank}(A) = 4$
- C. $\text{rank}(A) \geq 4$
- D. $\text{rank}(A) \leq 4$ ✓

EXPLANATION

$\text{rank}(A) \leq 4$ because the system has infinitely many solutions ($m \geq 1$).

50. Let us consider the quadratic form defined by $q(x, y) = (x, y)A(x, y)$ with $A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. Which of the following statements is true?

- A. The quadratic form $q(x, y)$ is negative definite.
- B. $q(a, b) > 0$ for every $(a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ ✓
- C. The matrix A has two eigenvalues of different sign.
- D. $q(x, y) = 0$ is the equation of two lines passing through the origin.

EXPLANATION

The quadratic form $q(x, y) = 3x^2 - 2xy + 2y^2$ is positive definite (eigenvalues 2 and 3), so $q(a, b) > 0$ for all $(a, b) \neq (0, 0)$.

51. In the euclidean 3-dimensional space with a fixed cartesian system, let us $C_1: x^2 + y^2 + z^2 - 1 = 0$, $C_2: x^2 + y^2 + z^2$ having radius respectively R_1 and R_2 . Which of the following statements is true?

- A. (a) $R_1 > R_2$
- B. (b) C_1 consists of only one point
- C. (c) $R_1 < R_2$
- D. (d) $R_1 = R_2$ ✓

EXPLANATION

Both C_1 and C_2 are spheres centered at $(0.5, 0.5, 0)$ with radius $\sqrt{1.5}$, so $R_1 = R_2$.

52. We are given a matrix $A \in \mathbb{R}^{n,n}$, symmetric and positive definite, and its Choleski factorization $A = R^T R$. Which one of the following statements is surely correct?

- A. the factorization always exists but it is not unique
- B. the uniqueness of the factorization depends on the rank of matrix A
- C. the factorization does not always exist, but when it exists it is unique
- D. the factorization always exists and it is unique ✓

EXPLANATION

Cholesky factorization of a symmetric positive definite matrix is always unique.

53. Consider the following set of Matlab commands: `x=linspace(-1,1,10); xx=linspace(-1,1); y=sin(pi * x); z=spline(x,y,xx);` What is their result?

- A. They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- B. They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- C. They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$
- D. They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$ ✓

EXPLANATION

`spline` computes and evaluates the not-a-knot cubic spline interpolant.

54. In exact arithmetic (base 10), the result of the multiplication $513.21 * 20.135$ represented in the normalized floating point notation is:

- A. $10.33348337 * 10^3$
- B. $0.10333 * 10^6$
- C. $0.1033348335 * 10^5$ ✓
- D. $1.0333 * 10^4$

EXPLANATION

$513.21 * 20.135 = 10333.48335 = 0.1033348335 * 10^5$.

55. In the Euclidean plane with fixed cartesian system Oxy , let C be the conic defined by the equation $5x^2 + y^2 - 10y + 25 = 0$. Which of the following statements is correct?

- A. C is a parabola.
- B. C is a hyperbola.
- C. None of the other statements is correct.
- D. C is a point. ✓

EXPLANATION

Completing the square gives $(x)^2 + (y - 5)^2 = 0$, so C is a point.

56. Given the sphere $S: x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1: x + y + z = 0$ $\pi_2: x + y + z = 1$ let us denote by R_i the radius of the circle $S \cap \pi_i$, and by C_i its center, then:

- A. $R_2 = R_1$
- B. C_1 and C_2 coincide
- C. $R_2 < R_1$ ✓
- D. $R_1 < R_2$

EXPLANATION

The distance from the sphere center to π_2 is greater than to π_1 , so $R_2 < R_1$.

57. Let $R_2[x]$ be the vector space of real polynomials of degree ≤ 2 . Let $f: \mathbb{R}^3 \rightarrow R_2[x]$ be a linear map such that $f(1,0,0) = x^2 + x$, $f(0,1,0) = x + 2$, $f(0,0,1) = 2x^2 + 4x + 4$. Which one of the following statements is true?

- A. $f(1,0,-1) = x^2 - 4$
- B. f is surjective
- C. $\dim(\ker f) = 2$
- D. $\dim(\ker f) = 1$ ✓

EXPLANATION

$\dim(\ker f) = 1$ because $\dim(\text{Im } f) = 2$ and $\dim(\mathbb{R}^3) = 3$.

58. Let us consider the line $r: \{ x = 2 + t, y = 2 - t, z = t \}$ Which of the following statements is true?

- A. There exist infinitely many planes containing r and the point $(2,2,2)$
- B. There exist infinitely many planes containing the point $(2,2,2)$ and orthogonal to r
- C. There exist infinitely many planes containing r and the points $(2,2,2)$ and $(2,2,0)$
- D. There exist infinitely many planes containing r and the point $(2,2,0)$ ✓

EXPLANATION

Infinitely many planes contain the line r and the point $(2,2,0)$ since the point lies on the line.

59. For all $t \in \mathbb{R}$, consider the triangle having vertices in $P_1 = (3,2,1)$, $P_2 = (t,1,1)$, $P_3 = (1,2,t)$. Which of the following statements is true? (a) The edges P_1P_2 and P_2P_3 are orthogonal for $t = 3$. (b) There is no value of t such that the edges P_1P_2 and P_1P_3 are orthogonal. (c) The edges P_1P_2 and P_1P_3 are orthogonal for at least one value of t . (d) The edges P_1P_2 and P_2P_3 are orthogonal for two different values of t .

- A. The edges P_1P_2 and P_2P_3 are orthogonal for $t = 3$.
- B. There is no value of t such that the edges P_1P_2 and P_1P_3 are orthogonal.
- C. The edges P_1P_2 and P_1P_3 are orthogonal for at least one value of t . ✓
- D. The edges P_1P_2 and P_2P_3 are orthogonal for two different values of t .

EXPLANATION

The edges P_1P_2 and P_1P_3 are orthogonal when $t = 2$.

60. Consider the matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 10 \end{pmatrix} \in \mathbb{R}^{3,3}$. Find the right statement.

- A. A is positive definite.
- B. A has negative eigenvalues.
- C. None of the other statements is correct. ✓
- D. A does not have 0 as an eigenvalue.

EXPLANATION

Eigenvalues are 0, 0, 10, so A is not positive definite.

61. Let A be a square matrix $n \times n$ with real coefficients such that $\det(A) \neq 0$ and $\det(A^2 + 2A) = 0$. Which of the following statements is correct?

- A. A has no real eigenvalue.
- B. -2 is an eigenvalue for A. ✓
- C. 0 is an eigenvalue for A
- D. None of the other statements is correct

EXPLANATION

$\det(A^2 + 2A) = 0$ implies -

62. Consider the following linear system with variables x_1, x_2, x_3, x_4 :

$\{x_1 + x_2 + x_3 = 0; x_2 + x_3 + x_4 = 0; 2x_1 + x_2 + 3x_3 = 0; x_1 - x_2 + x_3 - x_4 = 0\}$. Find the correct statement.

- A. The system has ∞^2 solutions.
- B. The system has exactly one solution.
- C. The system does not have solutions.
- D. The system has ∞^1 solutions. ✓

EXPLANATION

D) The system has ∞^1 solutions because the rank of the coefficient matrix is 2, so the solution space has dimension $4 - 2 = 2$, but the system is homogeneous, reducing it to ∞^1 solutions.

63. Consider the lines $r: \{x=1, y=3\}$ $s: \{x=-2, z=3\}$. Which of the following statements is true?

- A. There exists a plane containing r which is orthogonal to s ✓
- B. There is no plane containing s which is orthogonal to r
- C. There is no plane containing r which is parallel to s
- D. There exists a plane containing r and s

EXPLANATION

A, plane $x + y + z = 4$ contains r and is orthogonal to s (direction vector $(0,1,0) \cdot (1,0,0) = 0$).

64. Let $R_2[x]$ be the vector space of real polynomials of degree ≤ 2 . Let $f: R_2[x] \rightarrow R_2[x]$ be the endomorphism defined as: $f(ax^2 + bx + c) = 2cx^2 - ax + 3b$ and let A be the matrix associated to f with respect to the basis $B = (x^2, x, 1)$ (for the domain and codomain). Which one of the following statements is true?

- A. $A = \begin{pmatrix} 0 & 0 & -1 \\ 3 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$
- B. $A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 1 & 3 & 0 \end{pmatrix}$
- C. $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 3 & 0 \end{pmatrix}$
- D. $A = \begin{pmatrix} 0 & 0 & 2 \\ -1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$ ✓

EXPLANATION

The matrix A is $\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 3 & 0 \end{pmatrix}$ because $f(x^2) = 2x^2$, $f(x) = -x + 3$, $f(1) = -1$.

65. Consider the matrices $A = (1, 0, 3)$ and $B = (b_1, b_2, b_3)$. Find the correct statement.

- A. $A \cdot B$ is symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is not parallel to $(1, 0, 3) \in \mathbb{R}^3$
- B. $A \cdot B$ is symmetric for any $(b_1, b_2, b_3) \in \mathbb{R}^3$
- C. There does not exist $(b_1, b_2, b_3) \in \mathbb{R}^3$ such that $A \cdot B$ is symmetric
- D. $A \cdot B$ is symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is parallel to $(1, 0, 3) \in \mathbb{R}^3$ ✓

EXPLANATION

$A \cdot B$ is symmetric only if vectors are parallel (dot product is scalar, symmetric matrix requires outer product symmetry).

66. For all $t \in \mathbb{R}$, let us consider the points $P = (2, 2, 2)$, $Q = (0, 0, t)$, $R = (1, -1, t)$, $S = (-3, 2, 1+t)$. Which of the following statements is true?

- A. The points P, Q, R, S are vertices of a trapezoid for all values of t .
- B. The points P, Q, R, S are coplanar when $t = 7$.
- C. The points P, Q, R, S are vertices of a rectangle for all values of t .
- D. The points P, Q, R, S are coplanar for at least one value of $t \in \mathbb{R}$. ✓

EXPLANATION

The points are coplanar when the determinant of the matrix formed by vectors PQ, PR, PS is zero, which happens for at least one t (e.g., $t=7$).

67. Let $h \in \mathbb{R}$ and let U be the vector subspace of $\mathbb{R}^3$ generated by the vector $(0, 0, h)$. Let $V = \{(x, y, z) \in \mathbb{R}^3 : x + 3y + z = 0\}$. Which one of the following statements is true?

- A. (a) For $h = 0$ it holds that $U \cap V = \emptyset$
- B. (b) $\dim(U \cap V) = 1$ for any h
- C. (c) $\dim(U) = 1$ for any h
- D. (d) $\dim(U \cap V) = 0$ for any h ✓

EXPLANATION

$\dim(U \cap V) = 0$ for any h because U is 1D and V is 2D, intersecting only at the zero vector unless $h=0$.

68. Consider the sphere $S: x^2 + y^2 + z^2 - 2x + 2y = 7$. The circle $\pi \cap S$ has radius R and center C such that

- A. $C = (1, -1, 0)$, $R = 3$ ✓
- B. $C = (0, 0, 0)$, $R = 3$
- C. $C = (1, -2, 1)$, $R = 7$
- D. $C = (-1, 1, 0)$, $R = 3$

EXPLANATION

Completing the square: $(x-1)^2 + (y+1)^2 + z^2 = 9$, so center $C=(1,-1,0)$, radius $R=3$.

69. Let $R_3[x]$ be the vector space of polynomials in the variable x , of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $R_3[x]$ is a vector subspace?

- A.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b = 0\}$. ✓
- B.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ad - bc = 0\}$.
- C.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b + c + d = 1\}$.
- D.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ac = 0\}$.

EXPLANATION

The set $\{p(x) = ax^3 + bx^2 + cx + d : a + b = 0\}$ is closed under addition and scalar multiplication.

70. Let $R_3[x]$ be the vector space of polynomials in the variable x , of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $R_3[x]$ is a vector subspace?

(a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b = 0\}$. (b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ad - bc = 0\}$. (c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b + c + d = 1\}$. (d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ac = 0\}$.

- A.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b = 0\}$. ✓
- B.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ad - bc = 0\}$
- C.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : a + b + c + d = 1\}$
- D.** The set $\{p(x) = ax^3 + bx^2 + cx + d \in R_3[x] : ac = 0\}$

EXPLANATION

The set $\{p(x) = ax^3 + bx^2 + cx + d : a + b = 0\}$ is closed under addition and scalar multiplication.

71. Consider the matrix $A = \begin{pmatrix} -1 & -1 & 2 & 1 \\ 0 & 1 & 0 & -1 \\ -1 & 2 & 0 & 1 \end{pmatrix}$. Find the correct statement.

- A.** $A^{-1} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ -1 & -3 & 1 & -1 \\ -1 & -1 & 1 & 1 \end{pmatrix}$
- B.** $A^{-1} = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 1 & 2 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$
- C.** $A^{-1} = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 2 & 2 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$
- D.** $A^{-1} = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 2 & -3 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}$. ✓

EXPLANATION

D, $A^{-1} = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 2 & -3 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}$ satisfies $A \cdot A^{-1} = I$.

72. Let $\hat{z}_h, \check{z}_h, \tilde{z}_h$ the versors of the coordinate axes of $\mathbb{R}^3$. For every $a \in \mathbb{R}$, let us consider the vectors $\check{z}_u = -2\hat{z}_h + a\tilde{z}_h$, $\check{z}_u = 2\hat{z}_h + \tilde{z}_h$, $\check{z}_u = -a\hat{z}_h - \tilde{z}_h$. Which of the following statements is true?

- A.** There exists a unique $a \in \mathbb{R}$ such that the vectors $\check{z}_u, \check{z}_u, \check{z}_u$ are linearly independent.
- B.** For every $a \in \mathbb{R}$ the vectors $\check{z}_u, \check{z}_u, \check{z}_u$ are coplanar. ✓
- C.** For every $a \in \mathbb{R}$ the vectors $\check{z}_u, \check{z}_u, \check{z}_u$ are linearly independent.
- D.** There exists a unique $a \in \mathbb{R}$ such that the vectors $\check{z}_u, \check{z}_u, \check{z}_u$ are coplanar.

EXPLANATION

The vectors are always coplanar as their scalar triple product is zero for all a : $(-2)(1)(-a) + (a)(2)(-1) + (0)(1)(-1) = 0$.

73. Given the straight lines $r: \{2x - 5y = 7, 2z - 3y = 5\}$ and $s: \{x - y - z = 1, x - 4y + z = 5\}$, find the right statement.

- A. The lines are skew.
- B. The lines intersect at exactly one point.
- C. The lines are parallel and distinct. ✓
- D. None of the other statements is correct.

EXPLANATION

Direction vectors $(5, 2, 3)$ and $(3, 1, 1)$ are proportional, and no intersection exists (inconsistent system).

74. If $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, then

- A. $A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- B. $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ ✓
- C. $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- D. $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

EXPLANATION

$A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ because A is upper triangular with 1s on the diagonal.

75. In $\mathbb{R}^4$ let us consider the vector subspace $V = \{(x, y, z, w) \mid x + y - z - w = 0, x - 2y = 0\}$. Which one of the following statements is true? (a) V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, 1)$. (b) A basis of V is formed by 3 vectors. (c) V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$. (d) V is a subspace of dimension 1.

- A. V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, 1)$.
- B. A basis of V is formed by 3 vectors.
- C. V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$. ✓
- D. V is a subspace of dimension 1.

EXPLANATION

C) V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$ because these vectors satisfy the given equations and span the subspace.

76. In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic C having equation $x^2 - 6xy + 9y^2 + 2x - 6y + 1 = 0$. Which of the following statements is true?

- A. C is a not degenerate ellipse.
- B. C is degenerate. ✓
- C. C is a not degenerate hyperbola.
- D. C is a not degenerate parabola.

EXPLANATION

Discriminant $B^2 - 4AC = (-6)^2 - 4(1)(9) = 0$, indicating a degenerate conic (a repeated line).

77. Let us consider the quadratic form defined by $q(x, y) = (x, y)A(x, y)$ with $A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. Which of the following statements is true?

- A. $q(a, b) \geq 0$ for every $(a, b) \in \mathbb{R}^2$ ✓
- B. The matrix A has two negative eigenvalues
- C. There exists $(a, b) \in \mathbb{R}^2$ such that $q(a, b) < 0$
- D. The columns of A are eigenvectors for A

EXPLANATION

A) $q(a,b) \geq 0$ for every $(a,b) \in \mathbb{R}^2$ because the matrix A is positive semi-definite (all eigenvalues are non-negative).

Answer Key

1. D	2. B	3. A	4. B	5. A	6. A
7. A	8. A	9. C	10. C	11. C	12. B
13. A	14. C	15. A	16. A	17. A	18. A
19. B	20. B	21. A	22. A	23. D	24. C
25. A	26. A	27. A	28. A	29. B	30. B
31. D	32. B	33. B	34. B	35. D	36. A
37. B	38. D	39. B	40. B	41. C	42. D
43. D	44. C	45. D	46. B	47. B	48. B
49. D	50. B	51. D	52. D	53. D	54. C
55. D	56. C	57. D	58. D	59. C	60. C
61. B	62. D	63. A	64. D	65. D	66. D
67. D	68. A	69. A	70. A	71. D	72. B
73. C	74. B	75. C	76. B	77. A	